(10) Seat No.:________

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# SARDAR PATEL UNIVERSITY

## BCA (Semester – I) (CBCS) (Reg. & NC)

## US01CBCA02: Computer Organization

Monday, 20th March, 2017

**Time: 2.00pm To 5.00pm** **Max. Marks: 70**

**Note:** Figure indicates right side is maximum marks for each question

**Que: 1 Select an appropriate answer for the following.** **[10]**

1. Numbers are stored and transmitted inside a computer in ________ form
   **[A]** Binary **[B]** ASCII code **[C]** Decimal **[D]** Alphanumeric

2. Which statement is valid?
   **[A]** 1KB = 1024 bytes **[B]** 1 MB=2048 bytes
   **[C]** 1 MB = 1000 kilobytes **[D]** 1 KB = 1000 bytes

3. A byte corresponds to ________
   **[A]** 4 bits **[B]** 32 bits **[C]** 16 bits **[D]** 8 bits

4. ASCII stands for,
   **[A]** America standard coded information interchange
   **[B]** American standard code for information interchange
   **[C]** America standard coded interchange information
   **[D]** American standard coded interchange information

5. Extra bit added to a string of bits to detect errors is known as________
   **[A]** Additional bit **[B]** Correction bit **[C]** Parity bit **[D]** Updation bit

6. Which memory is permanent type of memory?
   **[A]** ROM **[B]** RAM **[C]** EPROM **[D]** EEPROM

7. Cache memory is being deployed for
   **[A]** Speed up rate of data fetching **[B]** Reduce rate of data fetching
   **[C]** To reduce the error **[D]** None of these

8. Floppy disk is a __________ disk.
   **[A]** Removable **[B]** Permanent **[C]** Direct access **[D]** None of these

9. Which one is a pointing device?
   **[A]** Scanner **[B]** Keyboard **[C]** Mouse **[D]** None of these

10. Which printer gives the best result of printing?
    **[A]** Dot-matrix printer **[B]** Laser printer
    **[C]** Inkjet printer **[D]** None of these

**Que: 2 Answer the following questions in brief: (Attempt any Ten)** **[20]**

1. Define : Software with examples
2. List applications of computer
3. Find Binary and Octal number equivalent to decimal 10
4. Explain ASCII character code with example
5. List steps of Instruction Execution cycle

6. Explain 1's complement method with example
7. Explain Cache memory
8. List stages of pipelining
9. Explain Static ROM
10. What is Scanner?
11. Give <u>two differences</u> between input device and output device
12. What is keyboard? List types of its keys.

**Que: 3 [A]** Draw and explain Block Diagram of Computer with its functions **[05]**

**[B]** Explain Hexadecimal Number System. Also Find $(ABC)_{16} = (?)_2$ **[05]**

**<u>OR</u>**

**Que: 3 [A]** List the generations of computers and explain any two in detail **[05]**

**[B]** Explain Binary addition and subtraction with suitable example **[05]**

**Que: 4 [A]** Explain Error detection and correction code (Even & Odd Parity bit) with example **[06]**

**[B]** Explain Excess Notation with example **[04]**

**<u>OR</u>**

**Que: 4 [A]** Explain Hamming code method with example **[06]**

**[B]** Write a note on CPU Organization (Vonn Neumann machine) **[04]**

**Que: 5 [A]** Explain pipelining machine through diagram **[05]**

**[B]** Explain array processor with diagram **[05]**

**<u>OR</u>**

**Que: 5 [A]** Explain Hard Disk with diagram. **[05]**

**[B]** Write a note on Registers **[05]**

**Que: 6** List all addressing techniques and explain any three of them in detail **[10]**

**<u>OR</u>**

**Que: 6** Explain Printer in detail with all its types **[10]**

## ### Best ☺f Luck ###

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